

Data Flow Diagram

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	2 Marks

The Data Flow Diagram below shows how weather data moves through the Rising Waters system, from user input to the predicted flood-risk output.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside the project that sends or receives data (e.g., User/Resident).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., 2.0 Preprocess Input).
Rectangle (solid fill)	Data Store	A place where data is held for later use (e.g., Trained Model file).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores.

Rising Waters Data Flow (Level 1)

Step	From	To	Data Passed
1.0	User (External Entity)	Web Form (Flask App)	Weather parameters: Temperature, Humidity, Cloud Cover, Rainfall figures
2.0	Web Form (Flask App)	Process: Validate & Convert Input	Raw form values
3.0	Process: Validate & Convert Input	Process: ML Prediction (Random Forest)	Numeric feature array (10 values)
4.0	Data Store: Trained Model (flood_model.pkl)	Process: ML Prediction	Loaded Random Forest model
5.0	Process: ML Prediction	Web Form (Flask App)	Predicted class (0 = No Flood, 1 = Flood)
6.0	Web Form (Flask App)	User (External Entity)	Displayed result: "No Flood" or "Flood Likely"